3. Scenario - pepper Noise

Scenario: A scanned document has small black and white dots scattered over it.

(a.) what is the noise called and what does it look like?

Salt and pepper noise is random black and white dots in an range. white pixel has value 255, and black pixels has value 0.

(b.) what problem does it cause?

1. Make it difficult to read
2. Reduce image Quality
3. cause error in ocr and Segmentation.

(c.) which filter removes it?

A median filter, usually with a 3x3 window, remove salt and pepper noise.

(d.) why is it better than an averaging filter?

The median filter removes extreme pixcels values while preserving edges. an averaging filter may blur the text and spread the effect of noise.

(e.) write your answer step by step.

1. Read the nosy image
2. Select a 3x3 pixels window
3. Sort the nine pixcel values
4. Select the middle value (median)
5. Replace the centre pixcel with the median.
6. Repeat for whole image.

Example: 12, 15, 14, 225, 13, 0, 16, 15, 14

11. **Low brightness Image :**

Scenario : An image taken in low light looks very dark and details are hard to see.

(a.) Why is the image so dark ?

The image is dark because it receives insufficient light during image capture. As a result, most pixels have low intensity values, making objects and background details difficult to distinguish.

(b.) What problems does it cause ?

1. Object become difficult to identify
2. Important details are hidden.
3. Contrast becomes a poor.
4. Segmentation and Edge detection become less accurate.

(c.) Which enhancement technique can be used ?

Histogram equalization can be used to improve the brightness distribution and contrast of a dark grayscale image.

(d.) Why did you choose it ?

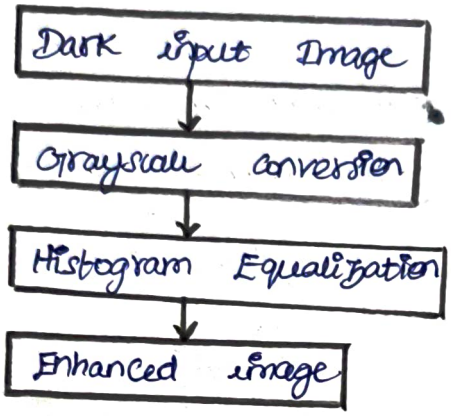
Histogram equalization redistributes pixel intensity values across a wider range. It can make hidden details more visible and improve the contrast between objects and background.

(e.) Write your answer step by step

1. Read the dark image.
2. Convert it to grayscale
3. Calculate the intensity histogram.
4. Apply histogram equalization

5. Generate the Enhancement image.
6. Compare the original and Enhancement images.

process diagram:



Histogram equalization improve image contrast and visibility, although missing details cannot always be received.

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18 Threshold selection problems:

scenario: An image needs segmentation, but choosing a suitable threshold is difficult.

(a.) why is choosing a threshold difficult?

Foreground and background pixels may have overlapping intensity values. noise and uneven illumination also make threshold selection difficult.

(b.) what issue occurs in threshold selection?

1. A low threshold may include background pixels.
2. A high threshold may remove object pixels.
3. Noise may create unwanted regions
4. Incorrect threshold produce poor segmentation.

(c.) which method can select it automatically?

Otsu's thresholding method can automatically select a suitable threshold based on image's grayscale histogram.

(d.) why did you choose it?

Otsu's method select a threshold that maximizes the between-class variance, helping separate background without manual selection.

(e.) Write your answer step by step.

1. Read the input image.
2. Convert it to Grayscale
3. Calculate the intensity histogram
4. Apply otsu's thresholding method.
5. Classify pixels into foreground and background.
6. Display and Evaluate the binary segmented image.

process diagram:

